

Final Report – SkillSwap App

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1. Project Aim and Description

The SkillSwap mobile application aims to connect individuals who want to share and exchange their skills with others. Users can list the skills they already have and the skills they want to learn. The platform provides personalized match recommendations, enabling users to find suitable partners for skill exchange, chat directly in-app, and even schedule sessions. The app promotes collaborative learning, reduces barriers to skill acquisition, and fosters a community of peer-to-peer learning.

Key features include:

- User authentication (sign up, login, and social login with Twitter/Facebook).
- Profile management (edit personal information, skills, and preferences).
- Skill-based matchmaking and history tracking.
- Real-time chat and notifications.
- Interactive map view for location-based filtering.
- Session booking and tracking.
- Search functionality and archived chats for better user experience.

2. Functional and Non-Functional Requirements

Functional Requirements:

1. Users must be able to sign up and sign in securely.
2. Users can add, edit, and delete their skills (both “Skills I Have” and “Skills I Want to Learn”).
3. The system must provide recommended matches based on skills.
4. Users can view matches on a map and filter by location.
5. Users can chat with matched partners through an integrated messaging system.
6. Users can edit their profile (name, email, bio, and location).
7. Users can view their match history and session history.
8. Users can book and manage learning sessions.
9. Notifications must be sent for new matches and messages.
10. Users can search for skills or people.
11. Users can access archived chats.

Non-Functional Requirements:

1. Usability: The UI must be simple, intuitive, and mobile-friendly.
2. Performance: Data retrieval and updates must complete within 2 seconds under normal network conditions.
3. Security: Only authenticated users can access the system. Firestore security rules ensure users can only modify their own data.
4. Scalability: The app should handle growth in the number of users and data without major architectural changes.
5. Reliability: The system should maintain 99% uptime and handle unexpected errors gracefully.
6. Compatibility: The app must run on both Android and iOS platforms.

3. User Stories

1. As a new user, I want to sign up using my email or social account so that I can start using the app quickly.
2. As a registered user, I want to edit my profile so that others can learn more about me.
3. As a learner, I want to add the skills I want to learn so I can find relevant matches.
4. As a mentor, I want to add the skills I have so I can share my knowledge with others.
5. As a user, I want to see recommended matches based on my skills so I can find suitable partners.
6. As a user, I want to chat with my matches so I can discuss skill exchange plans.
7. As a user, I want to see my past matches and sessions so I can track my learning progress.
8. As a user, I want to use the map to find matches near my location so that I can arrange in-person sessions.
9. As a user, I want to receive notifications for new messages or matches so I don't miss important updates.
10. As a user, I want to search for skills or people to find specific learning partners.
11. As a user, I want to move chats into archive so I can better organize my messages.

4. Test Cases

Feature	Test Case	Expected Result
Sign Up	Enter valid email and password, agree to terms	User is registered and redirected to Home screen.
Sign In	Enter valid email and password	User is logged in successfully.
Sign In	Enter invalid credentials	Error message is displayed.
Edit Profile	Update name, email, bio, and save	Data is saved and displayed correctly.
Add Skill	Enter a skill in 'Skills I Have' and save	Skill is added and displayed in the list.
Delete Skill	Remove an existing skill	Skill is removed from the list.
Match Recommendation	Add skills and view 'Recommended Matches'	Matches are displayed based on skill overlap.
Chat	Open chat with a matched user, send a message	Message appears in chat window.
Match History	Open match history	Past partners and sessions are displayed.
Map View	Open 'View on Map'	Matches are displayed on the map.
Session Booking	Schedule a new session	Session is saved and displayed in 'All Sessions'.
Delete Account	Press 'Delete Account'	User account is removed and app redirects to Sign In.
Search	Search for a skill or user name	Relevant results are displayed.
Archived Chats	Move a chat to archive	Chat disappears from active list and appears in archive.

5. Individual's Role and Responsibilities

Xingzuo Li:

- Implemented the following screens and functionalities:
 1. Sign Up
 2. Sign In
 3. Home Dashboard
 4. Edit Skills
 5. Recommended Matches
 6. Map View
 7. Edit Profile
- Integrated Firebase Authentication and Firestore data saving for profile and skills.

Joseph Wajsberg:

- Implemented the following screens and functionalities:
 1. Match History
 2. Chats (including archived chats)
 3. Sessions
 4. Settings
 5. Search functionality
 6. Notifications Manager implement (Firebase Messaging Service)
- Configured chat functionality, session management, and app-wide settings.

6. Task Distribution

Team Member

Tasks Completed

Xingzuo Li

Sign Up, Sign In, Home Dashboard, Edit Skills, Recommended Matches, Map View, Edit Profile

Joseph Wajsberg

Match History, Chats (including archived chats), Sessions, Settings, Search functionality, Notifications Manager implement (Firebase Messaging Service)

Deliverable 2 – Implementation Details

Based on the course outline, the following tasks were implemented and demonstrated in the SkillSwap app:

- Completed UI interface (Navigation flow of more than 10 screens: Sign Up, Sign In, Home, Edit Skills, Recommended Matches, Map View, Edit Profile, Match History, Chats, Sessions, Settings).
- API calls (Implemented with Firebase Firestore queries, asynchronous skill and chat data retrieval, and user profile updates).
- Notification manager (New match and new message notifications implemented using Firebase Cloud Messaging).
- Authentication (Email & Password authentication, with options for social login such as Twitter and Facebook).
- Location awareness (Google Maps integration with location pin and filter by location for matches).
- Running on an emulator (All functionalities tested and demonstrated on Android Emulator – Pixel 8a).